RAG Learning

Required Practicals — OCR B GCSE Chemistry

OCR GCSE Chemistry B (Twenty First Century Science) (J258) — Chapter C8 Practical Skills: PAG1–PAG8 apparatus, techniques, methods and analysis. Practical questions can appear on all four papers.

Chapter C8 of the specification requires you to complete **at least eight practical activities**, grouped into eight **Practical Activity Groups (PAG1–PAG8)**, so that you use every apparatus and technique listed below. Your school can use OCR's suggested activities or its own. There is no practical coursework mark: instead, **at least 15% of the marks** in the written papers test practical skills — describing methods, drawing and labelling apparatus, identifying variables, judging accuracy, precision and validity, processing results and suggesting improvements. This page follows the PAG order, using the example activity for each group plus the other practicals the specification links to it.

PAG	Title	Example activity in the specification	Linked content
PAG1	Reactivity trend	Using displacement reactions to identify the reactivity trend of Group 7 elements	C2.2, C3.2
PAG2	Electrolysis	Electrolysis of aqueous sodium chloride or aqueous copper sulfate, testing for the gases produced	C1.1, C1.4, C3.3
PAG3	Separation techniques	Using chromatography to identify mixtures of dyes in a sample of unknown composition	C3.4, C5.1
PAG4	Distillation	Distillation of a mixture, e.g. orange juice, cherry cola, hydrocarbons or inks	C5.1

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PAG5	Identification of species	Identifying an unknown compound using cation tests, anion tests and flame tests	C5.2
PAG6	Titration	Titrating a strong acid against a strong alkali to find the concentration of the acid, using a suitable indicator	C5.4
PAG7	Production of salts	Producing a pure, dry sample of an insoluble salt and a soluble salt	C5.1, C6.1
PAG8	Measuring rates of reaction	Investigating the effect of surface area, concentration and temperature on the rate of a reaction	C6.2

Exam Tip: The content chapters also tag separation work to more than one PAG: fractional distillation of crude oil (C3.4) and filtration, crystallisation and distillation (C5.1) are marked PAG3/PAG7, while chromatography (C5.1) is marked PAG4. Apparatus and techniques overlap between groups, so learn every technique — not just the example activity.

PAG1 — Reactivity Trend: Halogen Displacement Reactions (C2.2)

Apparatus and techniques: safe use and careful handling of gases, liquids and solids, including careful mixing of reagents under controlled conditions, using appropriate apparatus to explore chemical changes and/or products.

Aim: Use displacement reactions between the halogens and solutions of halide salts to put chlorine, bromine and iodine in order of reactivity.

- **Independent variable**: the halogen added and the halide solution it is added to.
- **Dependent variable**: whether a colour change (displacement) happens.
- **Controls**: same volume and concentration of each solution, same number of drops of halogen solution, viewed against a white background.

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Method:

1. Place a few drops of potassium chloride, potassium bromide and potassium iodide solution in separate wells of a spotting tile (or about 1 cm depth in test tubes).
2. Add a few drops of chlorine water to each. Record any colour change.
3. Repeat with fresh halide solutions, adding bromine water, then iodine solution.
4. A colour change shows that the added halogen has **displaced** a less reactive halogen from its compound.
5. The halogen that displaces the most others is the most reactive.

Halogen added	KCl(aq)	KBr(aq)	KI(aq)
Chlorine water (pale green)	—	turns orange — bromine formed	turns brown — iodine formed
Bromine water (orange)	no change	—	turns brown — iodine formed
Iodine solution (brown)	no change	no change	—

$$\text{chlorine} + \text{potassium bromide} \rightarrow \text{potassium chloride} + \text{bromine}$$

$$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$

Conclusion: reactivity order is chlorine > bromine > iodine — reactivity **decreases down Group 7**, because the outer shell is further from the nucleus, so the atom gains an electron less easily.

Other reactivity-trend practicals in the specification

- **Group 1 (C2.2, teacher demonstration):** small pieces of lithium, sodium and potassium are added to a trough of water behind a safety screen. Lithium fizzes steadily, sodium melts into a ball and whizzes around, potassium burns with a lilac flame. Each makes hydrogen and an alkaline hydroxide solution — reactivity **increases down Group 1**.

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- **Metals with water and dilute acid (C3.2):** add equal-sized pieces of different metals to the same volume and concentration of dilute hydrochloric acid and compare the rate of bubbling (or temperature rise). More vigorous bubbling = more reactive metal.
- **Metals with metal oxides (C3.2):** heat zinc, iron and copper each with the oxides of the other two. A reaction (glow, colour change) shows the metal is more reactive than the metal in the oxide, e.g. zinc + copper oxide → zinc oxide + copper.
- **(HT)** Displacement can be shown as an ionic equation: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$.

Safety: Chlorine and bromine are toxic and irritating to the eyes and lungs — use dilute solutions in small volumes in a well-ventilated room and wear eye protection. Group 1 metals are only handled by the teacher, using tiny pieces and a safety screen.

Exam Tip: Iodine solution and dilute bromine water can look similar. Use a white tile, and say what you *see* (e.g. "colourless solution turns orange"), not just "it reacted".

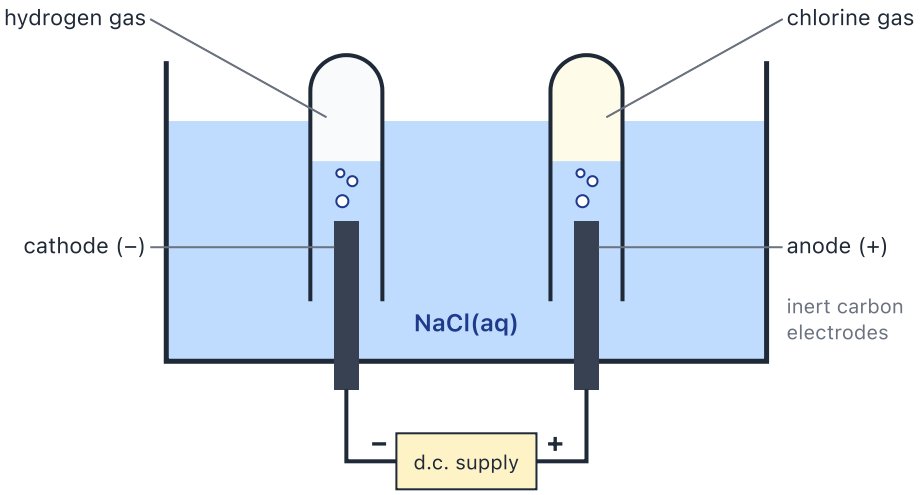
PAG2 — Electrolysis of Aqueous Solutions (C3.3)

Apparatus and techniques: use of appropriate apparatus and techniques to draw, set up and use electrochemical cells for separation and production of elements and compounds; use of appropriate qualitative reagents and techniques to analyse and identify products, including gas tests.

Aim: Electrolyse aqueous sodium chloride or aqueous copper sulfate with inert (carbon) electrodes and identify the product at each electrode.

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Electrolysis Cell with Gases Collected



Method:

1. Pour the electrolyte (e.g. concentrated sodium chloride solution) into the cell so that it covers the carbon electrodes.
2. Fill two small test tubes with the solution and invert one over each electrode, keeping them full.
3. Connect the electrodes to a low-voltage d.c. supply (about 6 V) and switch on. Observe each electrode.
4. When a test tube has collected gas, remove it carefully (keeping it upside down for hydrogen) and test the gas.
5. For copper sulfate solution, look for a pink-brown coating on the cathode and collect the gas from the anode.

Electrolyte (inert electrodes)	Cathode (−)	Anode (+)
Concentrated sodium chloride solution	hydrogen (sodium is more reactive than hydrogen)	chlorine (concentrated chloride ions)

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Copper sulfate solution	copper (less reactive than hydrogen)	oxygen (from hydroxide ions in the water)

Gas	Test	Positive result
Hydrogen	Hold a lighted splint at the mouth of the tube	Burns with a squeaky pop
Oxygen	Put a glowing splint into the tube	Splint relights
Carbon dioxide	Bubble the gas through limewater	Limewater turns milky (cloudy)
Chlorine	Hold damp blue litmus paper in the gas	Turns red, then is bleached white

(HT) Half equations: at the cathode, ions gain electrons (reduction), e.g. $2H^+ + 2e^- \rightarrow H_2$ or $Cu^{2+} + 2e^- \rightarrow Cu$. At the anode, ions lose electrons (oxidation), e.g. $2Cl^- \rightarrow Cl_2 + 2e^-$.

Extension investigations in the specification: investigate how the concentration of the sodium chloride solution, the current and the voltage affect the products and the amount of product (e.g. dilute sodium chloride gives some oxygen at the anode as well as chlorine). You can also test which substances are electrolytes by checking whether a lamp lights when solids, melts and solutions are placed in a simple circuit — ionic compounds conduct only when molten or dissolved, because the ions can then move.

Safety: Chlorine is toxic — run the electrolysis for only a short time, in a well-ventilated room, and do not smell the gas. Use a low-voltage supply. Copper sulfate solution is harmful — wear eye protection.

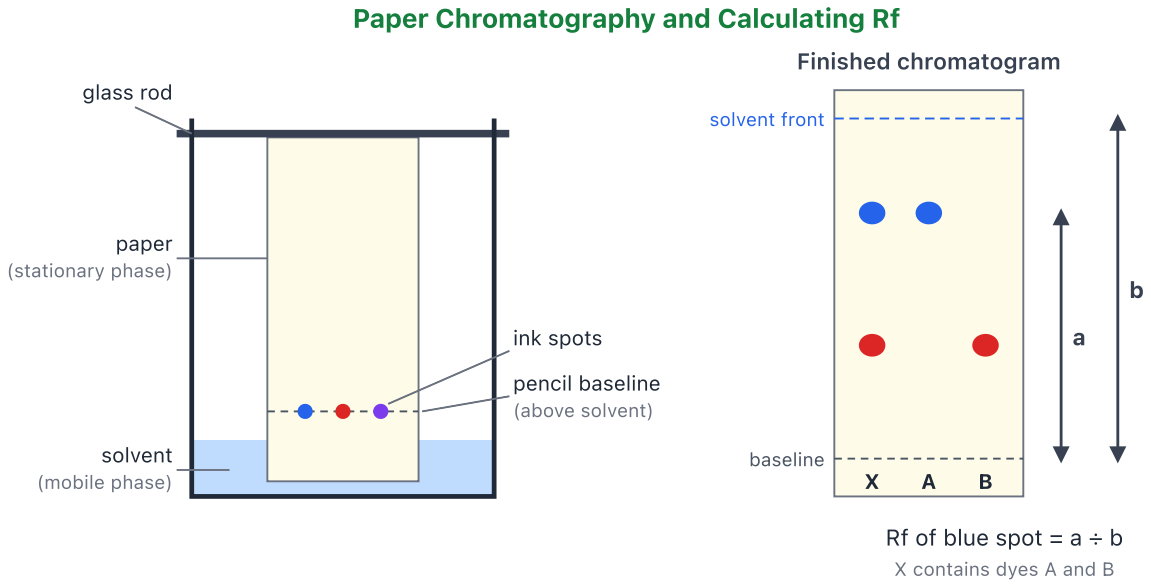
Exam Tip: Metals or hydrogen always form at the cathode; non-metals form at the anode. In aqueous solutions, hydrogen forms at the cathode unless the metal is *less* reactive than hydrogen (e.g. copper, silver).

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PAG3 — Separation Techniques: Chromatography of Dyes (C5.1)

Apparatus and techniques: safe use of a range of equipment to purify and/or separate chemical mixtures, including evaporation, filtration, crystallisation, chromatography and distillation.

Aim: Use paper chromatography to find out which dyes are in an unknown ink, by comparing it with known reference dyes.



Method:

1. Draw a baseline in **pencil** about 2 cm from the bottom of a strip of chromatography paper.
2. Put a small, concentrated spot of the unknown ink and of each reference dye on the line, spaced apart.
3. Hang the paper in a beaker containing solvent so that the solvent level is **below** the baseline. Cover with a lid to stop the solvent evaporating.
4. Remove the paper when the solvent is near the top. Mark the solvent front in pencil straight away and let the paper dry.
5. Compare spots: a dye in the unknown that travels the same distance as a reference dye (same Rf, same solvent) is probably the same substance. Calculate each Rf value.

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$$\text{Rf} = \text{distance moved by substance} \div \text{distance moved by solvent front}$$

- Chromatography uses a **stationary phase** (the paper) and a **mobile phase** (the solvent). Substances separate because they are distributed differently between the two phases: those that spend more time dissolved in the solvent move further.
- A **pure** substance gives one spot (in any solvent); a mixture gives more than one.
- Substances insoluble in water need a **non-aqueous solvent** (e.g. ethanol). Colourless substances need a **locating agent** sprayed on to make the spots visible.
- Rf is always less than 1 and has no units. It depends on the solvent, so compare with references run on the same paper in the same solvent.

Exam Tip: Use pencil for the baseline because ink would dissolve and run with the solvent. The baseline must be above the solvent so the spots do not dissolve into the solvent in the beaker.

Safety: Many chromatography solvents (e.g. ethanol) are flammable — keep them away from flames and use a lid.

Other separation techniques (C5.1, C3.4)

Technique	Separates	Example
Filtration	an insoluble solid from a liquid	sand from salt solution; excess base from a salt solution
Evaporation	a dissolved solid from its solvent (solvent lost)	salt from salt water
Crystallisation	a soluble solid as pure crystals	copper sulfate crystals from solution
Simple distillation	a solvent from a solution (solvent kept)	water from ink or sea water

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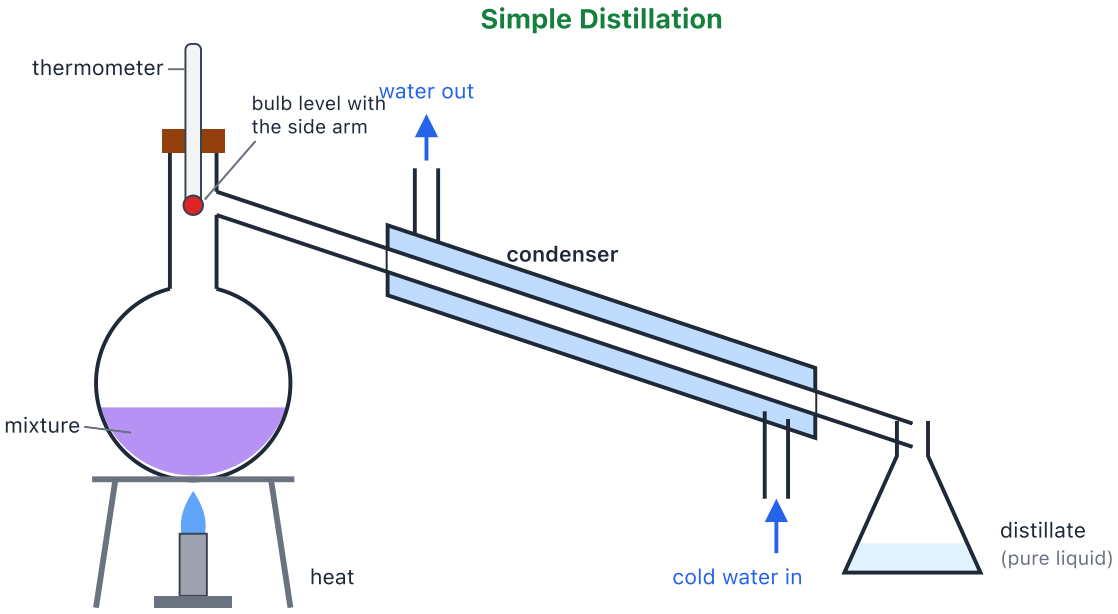
Fractional distillation	miscible liquids with different boiling points	ethanol and water; fractions of crude oil
Chromatography	dissolved substances with different solubilities	dyes in an ink

To choose a technique, look at the properties of the substances: state, solubility in the solvent and boiling point. Purity can be checked by measuring the **melting point** — a pure substance melts sharply at its data-book value; an impure one melts lower and over a range.

PAG4 — Distillation (C5.1)

Apparatus and techniques: safe use of a range of equipment to purify and/or separate chemical mixtures, including distillation; safe use of appropriate heating devices and techniques, including a Bunsen burner and a water bath or electric heater; use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature and volume of liquids and gases.

Aim: Separate a pure liquid from a mixture (e.g. water from ink, orange juice or cola) by simple distillation, and separate liquids with different boiling points by fractional distillation.



Method: Simple distillation (e.g. water from ink):

1. Put the mixture in a round-bottomed flask with a few anti-bumping granules. Fit a thermometer so its bulb is level with the side arm — it then reads the temperature of the vapour that is being distilled.
2. Connect the condenser so cold water enters at the **bottom** (lower end) and leaves at the **top** — the jacket stays full of water and cooling is most efficient.
3. Heat gently (Bunsen burner, or a water bath/electric heater for flammable liquids). The liquid with the lowest boiling point evaporates, the vapour is cooled in the condenser and condenses, and the pure liquid (distillate) collects in the receiver.
4. The temperature stays steady at the boiling point of the liquid being collected (100 °C for water). The dissolved solids (e.g. dyes) stay in the flask.

Fractional distillation (e.g. ethanol and water):

1. Fit a fractionating column (packed with glass beads) between the flask and the still head.
2. Heat slowly. The column is hotter at the bottom and cooler at the top; vapours condense and re-evaporate many times, so only the liquid with the lowest boiling point reaches the top first.
3. Collect fractions in separate receivers, changing the receiver each time the thermometer reading rises to a new steady value.

Fractional distillation of crude oil (C3.4) works on the same principle on an industrial scale: the oil is vaporised and fractions condense at different heights of the column, with the smallest molecules (lowest boiling points, weakest intermolecular forces) leaving at the top.

Safety: Never heat a sealed apparatus — the receiver must be open to the air. Use anti-bumping granules. Ethanol and hydrocarbons are flammable: heat with a water bath or electric heater, not a naked flame. Hot glassware looks the same as cold.

Exam Tip: Common diagram marks: thermometer bulb at the side arm (not in the liquid), water in at the bottom of the condenser, the apparatus open at the collecting end, and the heat source labelled.

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PAG5 — Identification of Species: Flame Tests and Ion Tests (C5.2)

Apparatus and techniques: use of appropriate qualitative reagents and techniques to analyse and identify unknown samples or products, including gas tests, flame tests and precipitation reactions; safe use of appropriate heating devices and techniques, including a Bunsen burner.

Aim: Identify the positive ion (cation) and negative ion (anion) in an unknown ionic compound.

Method: Flame test (metal ions):

1. Clean a nichrome wire loop by dipping it in dilute hydrochloric acid and holding it in a roaring (blue) Bunsen flame until no colour is seen.
2. Dip the loop in the acid, then into the solid sample.
3. Hold it at the edge of the blue flame and record the flame colour. Clean the loop between samples.

Sodium hydroxide test (metal ions in solution):

1. Dissolve a little of the sample in distilled water in a test tube.
2. Add a few drops of dilute sodium hydroxide solution and record the colour of any precipitate.
3. If the precipitate is white, add excess sodium hydroxide and see whether it dissolves.

Anion tests: add the reagents in the table below to fresh samples of the solution.

Metal ion	Flame colour	With sodium hydroxide solution
Lithium, Li^+	crimson (red)	—
Sodium, Na^+	yellow	—
Potassium, K^+	lilac	—
Calcium, Ca^{2+}	orange-red	white precipitate, does not dissolve in excess
Copper(II), Cu^{2+}	green (blue-green)	blue precipitate

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Iron(II), Fe^{2+}	—	green precipitate
Iron(III), Fe^{3+}	—	orange-brown precipitate
Zinc, Zn^{2+}	—	white precipitate, dissolves in excess

Anion	Test	Positive result
Carbonate, CO_3^{2-}	Add dilute acid; pass the gas through limewater	Fizzing; limewater turns milky (CO_2)
Chloride, Cl^-	Add dilute nitric acid, then silver nitrate solution	White precipitate
Bromide, Br^-	Add dilute nitric acid, then silver nitrate solution	Cream precipitate
Iodide, I^-	Add dilute nitric acid, then silver nitrate solution	Yellow precipitate
Sulfate, SO_4^{2-}	Add dilute hydrochloric acid, then barium chloride solution (or acidified barium nitrate)	White precipitate

$$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s) \quad Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s) \quad Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

- The acid is added first in the halide and sulfate tests to remove carbonate ions, which would also give a precipitate.
- Make sure the sample is **representative** of the whole material being analysed.
- **Instrumental methods** such as emission spectroscopy are more sensitive, accurate and faster than these lab tests, and can identify ions in mixtures where flame colours would mask each other (e.g. yellow sodium hides other colours) — but the equipment is expensive.

Safety: Wear eye protection. Sodium hydroxide is corrosive to eyes and skin. Silver nitrate stains skin; barium compounds are toxic. Take care with the hot wire loop and Bunsen flame.

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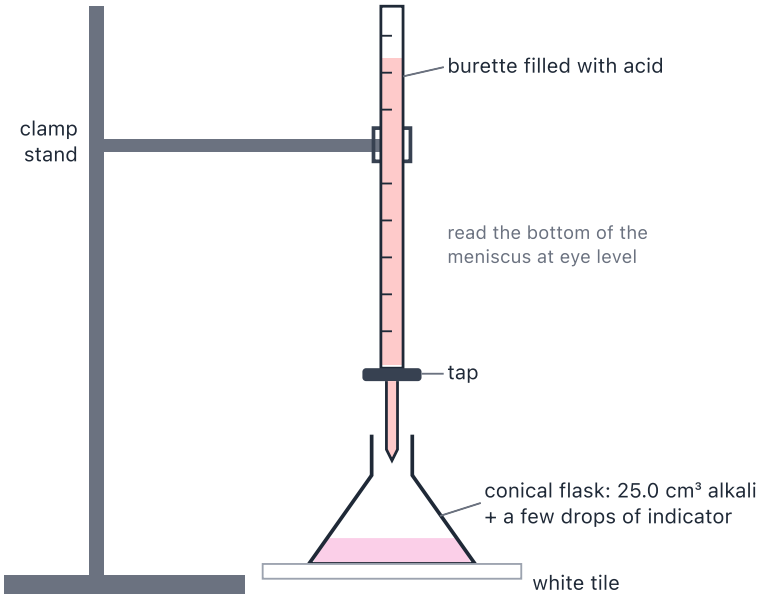
Exam Tip: Name the precipitate colour *and* the reagent. "White precipitate" alone is not enough to tell calcium from zinc — you must describe what happens with excess sodium hydroxide.

PAG6 — Titration: Finding the Concentration of an Acid (C5.4)

Apparatus and techniques: use of appropriate apparatus and techniques for conducting and monitoring chemical reactions, including appropriate reagents and/or techniques for the measurement of pH; determination of concentrations of strong acids and strong alkalis; making and recording measurements of volumes of liquids accurately.

Aim: Find the exact volume of a strong acid (e.g. hydrochloric acid) needed to neutralise a measured volume of a strong alkali (e.g. sodium hydroxide) of known concentration, and use it to work out the acid's concentration.

Titration Apparatus



Method:

1. Rinse a pipette with the alkali, then use it (with a pipette filler) to transfer exactly 25.0 cm³ of sodium hydroxide solution into a conical flask.

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2. Add a few drops of indicator (e.g. phenolphthalein, pink in alkali, or methyl orange, yellow in alkali). Stand the flask on a white tile.
3. Rinse the burette with the acid, fill it (using a funnel, then remove the funnel), run a little through the tap to fill the jet, and record the starting reading to 0.05 cm³.
4. Do a **rough titration**: add acid while swirling until the indicator changes colour. Record the final reading.
5. Repeat accurately: add acid quickly to within about 2 cm³ of the rough titre, then **drop by drop**, swirling, until a single drop changes the colour (phenolphthalein: pink → colourless; methyl orange: yellow → orange/red). This is the end point.
6. Repeat until you have **concordant** titres (within 0.10 cm³ of each other). Calculate the mean, ignoring the rough titre and any anomalous results.

	Rough	Run 1	Run 2	Run 3
Final reading / cm³	24.20	23.75	47.40	23.80
Initial reading / cm³	0.00	0.00	23.75	0.10
Titre / cm³	24.20	**23.75**	**23.65**	**23.70**

Mean of concordant titres = (23.75 + 23.65 + 23.70) ÷ 3 = **23.70 cm³**.

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l) \qquad \text{ionic: } H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

(HT) Calculating the concentration: if the sodium hydroxide is 0.100 mol/dm³:

- moles of NaOH = 0.100 × (25.0 ÷ 1000) = 0.00250 mol
- the reaction is 1 : 1, so moles of HCl = 0.00250 mol
- concentration of HCl = 0.00250 ÷ (23.70 ÷ 1000) = **0.105 mol/dm³**

$$\text{(HT) concentration (mol/dm}^3\text{)} = \text{number of moles of solute} \div \text{volume (dm}^3\text{)} \qquad \text{concentration (g/dm}^3\text{)} = \text{mass of solute (g)} \div \text{volume (dm}^3\text{)}$$

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Making a standard solution: weigh the solid accurately, dissolve it in a beaker of distilled water, transfer the solution and all the rinsings into a volumetric flask, make up to the mark with distilled water (bottom of the meniscus on the line), stopper and invert to mix. **Measuring pH** during neutralisation can be done with universal indicator or, more precisely, a pH meter.

- **Precise:** concordant repeat titres with a narrow range.
- **Accurate:** close to the true value — use a pipette and burette (not a measuring cylinder), read the meniscus at eye level, add drop by drop near the end point.
- **Valid:** measures what it should — e.g. the end point is a single-drop colour change, and the same indicator is used each time.

Safety: Sodium hydroxide and hydrochloric acid are irritant/corrosive even when dilute — wear eye protection. Always use a pipette filler. Fill the burette below eye level.

Exam Tip: Record burette readings to 2 decimal places (ending in 0 or 5). Titres should be "concordant" — say this word. Use a white tile so the colour change is easy to see, and swirl so the reagents mix.

PAG7 — Production of Salts (C6.1)

Apparatus and techniques: safe use of equipment to purify and/or separate mixtures, including evaporation, filtration and crystallisation; accurate measurement of mass and volume; safe handling and careful mixing of reagents; safe use of heating devices including a water bath or Bunsen burner; measuring pH.

Aim: Make a pure, dry sample of a soluble salt (copper sulfate) from an insoluble base, and a pure, dry sample of an insoluble salt (calcium carbonate) by precipitation.

Method: Soluble salt — copper sulfate crystals:

1. Warm about 25 cm^3 of dilute sulfuric acid in a beaker (water bath or gentle heating).
2. Add copper(II) oxide a spatula at a time, stirring, until no more dissolves and some black solid remains — the base is now in **excess**, so all the acid has reacted.

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3. **Filter** to remove the excess copper oxide. The filtrate is blue copper sulfate solution.
4. Heat the filtrate in an evaporating basin over a water bath to evaporate some of the water until crystals start to form at the edge.
5. Leave to cool so that crystals form slowly (**crystallisation**). Filter off the crystals and dry them between filter papers or in a warm oven.

Insoluble salt — calcium carbonate:

1. Mix solutions of calcium chloride and sodium carbonate in a beaker. A white precipitate of calcium carbonate forms.
2. Filter the mixture; the precipitate stays on the filter paper.
3. Wash the precipitate with distilled water to remove the soluble sodium chloride, then leave it to dry in a warm place.

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

$$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaCl(aq)$$

- Acids make salts with metals (salt + hydrogen), metal oxides and hydroxides (salt + water) and carbonates (salt + water + carbon dioxide).
- For a soluble salt from an alkali (no solid excess to filter off), first do a titration to find the exact volumes, then repeat without indicator — this is how ammonium salts for fertilisers can be made in the lab (C6.4).
- Calculate the **percentage yield** = actual mass ÷ theoretical mass × 100. Yields are below 100% because of loss during filtering and transferring, solution left in the filter paper, and incomplete crystallisation.

Safety: Wear eye protection. Do not boil the solution to dryness — it can spit, and copper sulfate crystals lose their water of crystallisation. Copper compounds are harmful. Take care with hot evaporating basins.

Exam Tip: Explain *why* each step is done: excess base makes sure all the acid is used up; filtering removes the unreacted base; slow cooling gives larger crystals; washing an

insoluble salt removes soluble impurities.

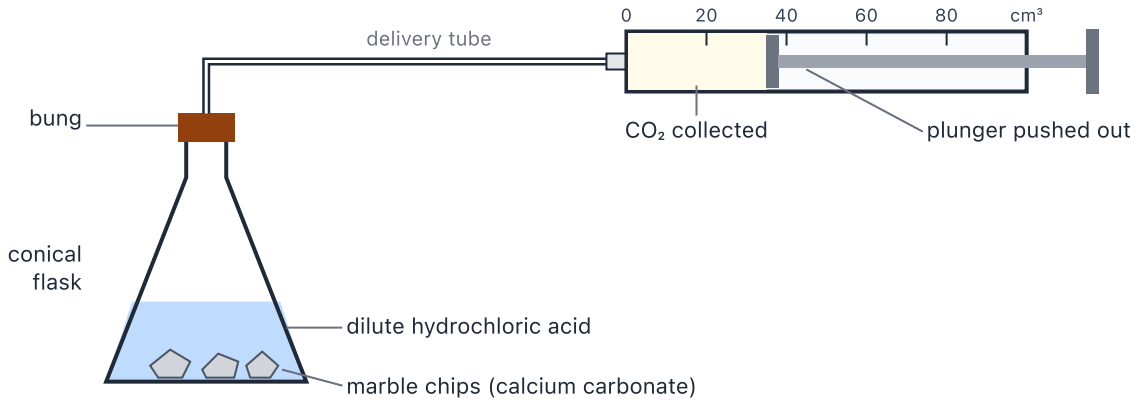
PAG8 — Measuring Rates of Reaction (C6.2)

Apparatus and techniques: use of appropriate apparatus to make and record measurements accurately, including mass, time, temperature and volume of liquids and gases; making and recording appropriate observations during chemical reactions, including changes in temperature and the measurement of rates of reaction by a variety of methods such as production of gas and colour change.

Aim: Investigate how surface area, concentration and temperature affect the rate of the reaction between calcium carbonate (marble chips) and hydrochloric acid.

- **Independent variable**: size of marble chips (surface area), concentration of acid, or temperature.
- **Dependent variable**: volume of carbon dioxide produced over time (or mass lost).
- **Controls**: mass of marble chips, volume of acid, and the two factors not being tested.

Measuring the Volume of Gas with a Gas Syringe



Method: Gas collection (gas syringe or collection over water):

1. Measure a fixed volume of hydrochloric acid into a conical flask and weigh a fixed mass of marble chips.
2. Add the chips, immediately insert the bung connected to the gas syringe and start the stopwatch.

3. Record the volume of gas every 30 seconds until no more gas is produced.
4. Repeat with different chip sizes (or acid concentrations, or temperatures using a water bath). Repeat each and calculate means.

Mass loss:

1. Put the flask of acid on a balance with a cotton-wool plug in the neck (lets CO_2 out but stops acid spray escaping).
2. Add the chips, replace the plug and record the mass every 30 seconds. The mass falls as carbon dioxide escapes.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Rate from a colour change or a precipitate (HT)

Some reactions do not give off a gas. If a reaction forms a precipitate or changes colour, its rate can be followed by timing the change by eye or with a colorimeter. In the "disappearing cross" experiment, sodium thiosulfate solution reacts with hydrochloric acid to form a yellow precipitate of sulfur; time how long it takes for a cross drawn under the flask to disappear when viewed from above. A shorter time means a faster rate (rate $\propto$ 1 ÷ time). Changing the temperature using a water bath, or diluting the thiosulfate with water while keeping the total volume the same, shows the effect of temperature and concentration.

Analysing the results

- Plot volume of gas (or mass lost) against time. The curve is steepest at the start (fastest rate) and levels off when a reactant is used up.
- **Mean rate** = amount of product formed ÷ time taken. The rate at a particular time is the **gradient of a tangent** to the curve at that time.
- Smaller chips (larger surface area to volume ratio), more concentrated acid or a higher temperature give a steeper initial curve. If the same masses and amounts of reactant are used, the final volume of gas is the same.
- Explain with collision theory: more frequent collisions (surface area, concentration) and, for temperature, collisions with more energy, so a greater proportion exceed the activation energy.

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Safety: Wear eye protection — acid can spray. The disappearing cross reaction produces sulfur dioxide, which can trigger asthma: use a well-ventilated room and dilute concentrations. Take care with hot water baths.

Exam Tip: A gas syringe is more precise than collecting over water in an upturned measuring cylinder. Errors come from losing gas before the bung is in — and carbon dioxide is slightly soluble in water, so collection over water gives low readings.

Other Practical Work (not a PAG)

The specification also suggests these practicals in the content chapters. They are not one of the eight PAGs, but questions can be set on them.

Practical	Chapter	Key points
Cooling curve for stearic acid (or heating curve for ice)	C1.1	Record temperature every 30 s; the flat section on the graph is the melting point, where energy is transferred without a temperature change while the state changes.
Exothermic or endothermic?	C1.2	Mix reactants in a polystyrene cup with a lid and record the temperature change with a thermometer. Temperature rise = exothermic; fall = endothermic. The cup reduces energy transfer to the surroundings.
Identifying unknown gases	C1.1, C1.4	Use the gas tests for hydrogen, oxygen, carbon dioxide and chlorine (see PAG2).
Properties of ionic compounds and other materials	C2.3, C4.1, C4.3	Test melting point, solubility, electrical conductivity (solid, molten, dissolved), hardness and flexibility to classify substances as ionic, simple molecular, giant covalent, metallic or polymer.
Transition metal compounds	C2.5	Observe the colours of transition metal compounds and compare how well they catalyse a reaction (e.g. the

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		decomposition of hydrogen peroxide, measuring the volume of oxygen produced).
Reactions of alkanes, alkenes and alcohols; cracking	C3.4, C4.2	Alkenes decolourise bromine water (orange → colourless) but alkanes do not; crack a long-chain hydrocarbon over a hot catalyst and test the gas for unsaturation with bromine water.
Rusting and corrosion prevention	C4.5	Set up iron nails in tubes with air and water, air only (drying agent) and water only (boiled water, oil layer); only the nail with both oxygen and water rusts. Compare barrier coatings and sacrificial protection (e.g. a nail joined to zinc or magnesium).
Reversible reactions	C6.3	e.g. heating blue hydrated copper sulfate gives white anhydrous copper sulfate; adding water reverses the change (exothermic).

Practical Skills — Everything You Need to Know (C8 and Ideas about Science)

The DfE apparatus and techniques that the PAGs must cover between them:

Apparatus and technique	Main PAGs
Accurate measurement of mass, time, temperature and volume of liquids and gases	PAG4, PAG6, PAG7, PAG8
Safe use of heating devices (Bunsen burner, water bath, electric heater)	PAG4, PAG5, PAG7
Safe handling and careful mixing of gases, liquids and solids to explore chemical changes	PAG1, PAG7
Purifying and separating mixtures: evaporation, filtration, crystallisation, chromatography, distillation	PAG3, PAG4, PAG7

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Conducting and monitoring reactions, including measuring pH	PAG6, PAG7
Observing changes during reactions, including temperature changes and measuring rates (gas production, colour change)	PAG8
Setting up and using electrochemical cells	PAG2
Qualitative tests (gas tests, flame tests, precipitation) and determining concentrations of strong acids and alkalis	PAG2, PAG5, PAG6

Skill	What examiners expect
Hypothesis and prediction	A hypothesis is a tentative explanation; a prediction says what will happen when a factor is changed (in words or as a sketch graph).
Variables	Identify the independent, dependent and control variables, and say how each control is kept the same.
Choice of apparatus	Justify it by precision and accuracy — e.g. burette and pipette rather than measuring cylinder; gas syringe rather than collection over water.
Range and sample size	Suggest a suitable range of values (at least five) and number of repeats, and justify it.
Risk assessment	Name the hazard (e.g. corrosive, toxic, flammable), the risk, and a sensible way to reduce it.
Accuracy	How close a result is to the true value.
Precision	How close repeated readings are to each other (small range = precise).
Repeatability	Same person, same method, same equipment → similar results.
Reproducibility	Different person or different method/equipment → similar results.
Random and systematic error	Random errors cause scatter (reduce by repeating and averaging); systematic errors shift every reading the same way (e.g. a balance not zeroed, heat loss).

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Outliers	Treat as data unless there is a reason to reject them (e.g. a measuring or recording error); explain your decision.
Processing data	Use SI units, prefixes and an appropriate number of significant figures; calculate mean and range; the true value probably lies within the range.
Graphs	Choose the right graph type, label axes with units, plot accurately, draw a line or curve of best fit, show range bars, and use gradients, interpolation and extrapolation.
Conclusions and evaluation	Describe the pattern, explain it with science, say how far the data supports the prediction, and suggest improvements and further investigations.

Exam Tip: **Common mark-scheme hooks:** "repeat and calculate a mean", "concordant titres", "use a white tile", "thermometer bulb level with the side arm", "pencil baseline above the solvent", "add excess base then filter", "measure with a gas syringe", "control the temperature with a water bath", "wear eye protection because … is corrosive".

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